HW #2 Solutions

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CSEE W3827 - Fundamentals of Computer Systems Spring 2026

Note that this homework has 8 problems and is 7 pages long.

1. Simplify the following Boolean expressions to a minimum number of literals:

   (a) $\bar{B}C + \bar{B}\bar{C}D$

   Answer: $= \bar{B}(C + \bar{C}D) = \bar{B}(C + D)$

   (b) $\overline{(\bar{Y} + Z)}(Y + \bar{Z})$

   Answer: $= Y\bar{Z}(Y + \bar{Z}) = Y\bar{Z} + Y\bar{Z} = Y\bar{Z}$

   (c) $\bar{C}\bar{D}A + C\bar{D}A + DA$

   Answer: $= \bar{D}(\bar{C}A + CA) + DA = A\bar{D}(C + \bar{C}) + DA = A\bar{D} + DA = A(\bar{D} + D) = A$

   (d) $X(\bar{C}Y + \bar{C}\bar{Y}) + X(\bar{W}\bar{C} + \bar{W}C)$

   Answer: $= X(\bar{C}(Y + \bar{Y})) + X(\bar{W}(C + \bar{C})) = X\bar{C} + X\bar{W} = X(\bar{C} + \bar{W})$

   (e) $(\bar{B}\bar{C} + BD + \bar{C}D)(\bar{B} + C + \bar{B}C)$

   Answer: $= \bar{B}\bar{C} + B\bar{B}D + \bar{B}\bar{C}D + \bar{B}C\bar{C} + CBD + C\bar{C}D + \bar{B}C\bar{C} + B\bar{B}CD + \bar{B}C\bar{C}D$
   $= \bar{B}\bar{C} + \bar{B}\bar{C}D + CBD = \bar{B}\bar{C}(1 + D) + CBD = \bar{B}\bar{C} + CDB$

2. Reduce the following Boolean expressions to the indicated number of literals:

   (a) $Y + \bar{Z}(W + \overline{Y + W})$ to two literals

   Answer: $= Y + \bar{Z}(W + \bar{W}\bar{Y}) = Y + \bar{Z}(W + \bar{Y}) = Y + W\bar{Z} + \bar{Y}\bar{Z} = W\bar{Z} + Y + \bar{Z}$
   $= \bar{Z}(W + 1) + Y = \bar{Z} + Y$

   (b) $\bar{Y}X + Y\bar{X}Z + \bar{Y}\bar{X}$ to three literals

   Answer: $= \bar{Y}(X + \bar{X}) + \bar{X}YZ = \bar{Y} + \bar{X}YZ = \bar{Y} + \bar{X}Z$

   (c) $\bar{Y}X(\bar{W} + ZW) + X(Y + \bar{Y}\bar{Z}W)$ to one literal

   Answer: $= \bar{W}X\bar{Y} + WX\bar{Y}Z + XY + WX\bar{Y}\bar{Z} = \bar{W}X\bar{Y} + WX\bar{Y}(Z + \bar{Z}) + XY$
   $= \bar{W}X\bar{Y} + WX\bar{Y} + XY = X\bar{Y}(W + \bar{W}) + XY = X\bar{Y} + XY = X(Y + \bar{Y}) = X$

3. Using DeMorgan's theorem (as many times as necessary), express the function $F = \bar{X}Z + X\bar{Z}Y + \bar{X}\bar{Y}$ with only

   (a) OR and complement operations

   Answer: $= \overline{X + \bar{Z}} + \overline{\bar{X} + Z + \bar{Y}} + \overline{X + Y}$

   (b) AND and complement operations

   Answer: $\overline{\overline{\bar{X}Z} \cdot \overline{X\bar{Z}Y} \cdot \overline{\bar{X}\bar{Y}}}$

4. Find the complement of the following expressions:

(a) $\bar{B}\bar{D} + BD$

Answer: In general, we simply need to flip each literal, change each original "AND" to "OR", and change each original "OR" to "AND" (but make sure to add parentheses to keep order of preference the same). So for this first problem we get that the complement is $(B + D)(\bar{B} + \bar{D})$.

(b) $(C + \bar{B}D)(C + \bar{B} + \bar{D})(\bar{C}\bar{B} + \bar{D})$

Answer: $\bar{C}(B + \bar{D}) + \bar{C}BD + (C + B)D$

(c) $\bar{W}\bar{Y}(Z\bar{X} + \bar{Z}X) + WY(Z + X)(\bar{Z} + \bar{X})$

Answer: $(W + Y + (\bar{Z} + X)(Z + \bar{X}))(\bar{W} + \bar{Y} + \bar{Z}\bar{X} + ZX)$.

(d) $(\bar{B} + \bar{D})AC + E$

Answer: $(BD + \bar{A} + \bar{C})\bar{E}$.

5. Convert the following into sum-of-products and product-of-sums forms:

(a) $(\bar{A} + \bar{B}\bar{D})(\bar{D} + A\bar{C})$

Answer: The easiest way to get an expression F into SoP form is to just "multiply out" and simplify. For PoS form is to do the following:

i. Compute F' (Swap AND for OR, OR for AND, flip literals)

ii. Write F' in simplified SoP form ("multiply out" and simplify)

iii. Compute $(F')'$ doing the swap and flip, and now you have F in PoS form

If $F = (\bar{A}+\bar{B}\bar{D})(\bar{D}+A\bar{C})$, then by multiplying out, we get $F = \bar{A}\bar{D}+\bar{A}A\bar{C}+\bar{B}\bar{D}\bar{D}+\bar{B}\bar{D}A\bar{C}$. The term $\bar{A}A\bar{C} = 0$ and $\bar{B}\bar{D}\bar{D} = \bar{B}\bar{D}$ and $\bar{B}\bar{D}A\bar{C}$ is subsumed within $\bar{B}\bar{D}$,, making the SoP solution equal to = $\bar{A}\bar{D} + \bar{B}\bar{D}$.
For PoS, if $F = \bar{A}\bar{D}+\bar{B}\bar{D}$, then $F' = (A+D)(B+D)$, multiplying out yields $AB+AD+BD+D = AB + D$, and re-complementing yields $F = (\bar{A} + \bar{B})\bar{D}$.

(b) $(\bar{Z} + Y)\bar{X}(\bar{X} + Z) + X$.

Answer: For SoP form, multiplying out yields $F = \bar{Z}\bar{X}\bar{X} + \bar{Z}\bar{X}Z + Y\bar{X}\bar{X} + Y\bar{X}Z + X = \bar{Z}\bar{X} + Y\bar{X} + X$.
For PoS, $F' = (Z + X)(\bar{Y} + X)\bar{X} = Z\bar{Y}\bar{X} + ZX\bar{X} + X\bar{Y}\bar{X} + XX\bar{X} = Z\bar{Y}\bar{X}$, so $F = \bar{Z} + Y + X$.

(c) $(\bar{Z} + XW + \bar{W}Y)(\bar{X} + \bar{Z}\bar{Y})$

Answer: For SoP, multiplying out yields $\bar{Z}\bar{X} + \bar{Z}\bar{Z}\bar{Y} + XW\bar{X} + XW\bar{Z}\bar{Y} + \bar{W}Y\bar{X} + \bar{W}Y\bar{Z}\bar{Y} = \bar{Z}\bar{X} + \bar{Z}\bar{Y} + \bar{W}Y\bar{X}$ (we noted some terms equaled 0 and that $XW\bar{Z}\bar{Y}$ was subsumed by $\bar{Z}\bar{Y}$).
For PoS, if $F = \bar{Z}\bar{X} + \bar{Z}\bar{Y} + \bar{W}Y\bar{X}$, then $F' = (Z + X)(Z + Y)(W + \bar{Y} + X)$, which, by multiplying out, is $ZZW + ZZ\bar{Y} + ZZX + ZYW + ZY\bar{Y} + ZYX + XZW + XZ\bar{Y} + XZX + XYW + XY\bar{Y} + XYX = ZW + Z\bar{Y} + ZX + XY$ (note that every 3-literal product term is subsumed by one of the 2-literal product terms in the expression). So $F = (\bar{Z} + \bar{W})(\bar{Z} + Y)(\bar{Z} + \bar{X})(\bar{X} + \bar{Y})$.

6. Simplify the following boolean expressions using a K-map:

(a) $B\bar{C} + CD + B\bar{D} + \bar{B}\bar{C}D$

Answer:

B \ CD	00	01	11	10
0	0	1	1	0
1	1	1	1	1

Simplified expression: $B + D$

(b) $\bar{Y}\bar{Z} + \bar{X}\bar{Z} + ZX\bar{Y}$

Answer:

X \ YZ	00	01	11	10
0	1	0	0	1
1	1	1	0	0

Simplified expression: $X\bar{Y} + \bar{X}\bar{Z}$

(c) $XW + \bar{X}\bar{Z} + X\bar{Z}\bar{W}$

Answer:

W \ XZ	00	01	11	10
0	1	0	0	1
1	1	0	1	1

Simplified expression: $\bar{Z} + XW$

7. Simplify in Sum-of-product form via a K-map (indicate what you identify as prime implicants and essential prime implicants). Recall that $m(i)$ is the product term whose variables are complemented when and only when their position corresponds to a '0' in the binary representation of i, e.g., $m(5) = \bar{W}X\bar{Y}Z$.

   (a) $F(W, X, Y, Z) = \sum m(0, 4, 5, 7, 9, 12, 13, 14)$

   Answer:

WX \ YZ	00	01	11	10
00	1	0	0	0
01	1	1	1	0
11	1	1	0	1
10	0	1	0	0

   PI that is not essential shown in gray. Others are EPIs. The EPIs completely cover so the (grey) PI, $X\bar{Y}$, should not be included
   $\bar{W}\bar{Y}\bar{Z} + \bar{W}XZ + WX\bar{Z} + W\bar{Y}Z$.

   (b) $F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 8, 9, 10, 11, 13, 15)$

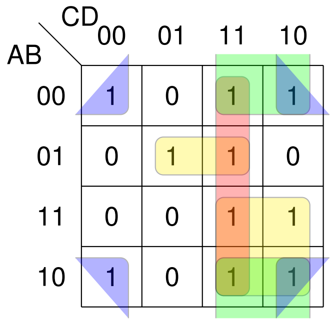
   Answer:

AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	0
11	0	1	1	0
10	1	1	1	1

   non-essential PIs ($\bar{C}D$ and $A\bar{B}$) are red. Others are EPIs. EPIs completely cover, so non-essential PIs should not be included.
   $\bar{A}\bar{C} + AD + \bar{B}\bar{D}$

   (c) $F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$

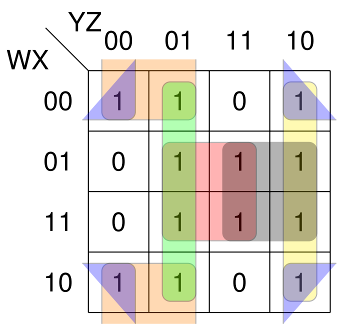
Answer:



non-essential PIs are red (CD) and green ($\bar{B}C$). EPIs shown in yellow and blue. Note that to cover $\bar{B}CD$ (i.e., the boxes that are third from left in top and bottom rows), one but only one of these non-essentials is needed.
$AC + \bar{A}BD + \bar{B}\bar{D} + CD$ or $AC + \bar{A}BD + \bar{B}\bar{D} + \bar{B}C$

(d) $F(W,X,Y,Z) = \sum m(0,1,2,5,6,7,8,9,10,13,14,15)$

Answer:



There are no EPIs. The following are all acceptable (obtained by selection rule) - there are also others. Although the first two are the best as they minimize the number of product terms.

- $XZ + \bar{X}\bar{Y} + Y\bar{Z}$
- $\bar{X}\bar{Z} + XY + \bar{Y}Z$
- $\bar{X}\bar{Z} + XZ + \bar{Y}Z + Y\bar{Z}$
- $\bar{Y}Z + Y\bar{Z} + XY + \bar{X}\bar{Y}$

(e) $F(W,X,Y,Z) = \sum m(0,1,2,5,8,9,11,12)$

Answer:

WX \ YZ	00	01	11	10
00	1	1	0	1
01	0	1	0	0
11	1	0	0	0
10	1	1	1	0

non-essential PI shown in blue. Others are EPIs. Non-essential PI should not be included.
$W\bar{Y}\bar{Z} + W\bar{X}Z + \bar{W}\bar{Y}Z + \bar{W}\bar{X}\bar{Z}$

(f) $F(A, B, C, D) = \sum m(1, 3, 6, 7, 9, 13, 14, 15)$

Answer:

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	0	1	1
11	0	1	1	1
10	0	1	0	0

EPI shown in yellow. Applying selection rule after choosing yellow, will choose green, red or grey next.
Choosing green or red first will yield $BC + \bar{A}\bar{B}D + A\bar{C}D$.
If choose grey first, then might wind up with $BC + \bar{B}\bar{C}D + ABD + \bar{A}CD$ or $BC + \bar{B}\bar{C}D + ABD + \bar{A}\bar{B}D$. There are also other possibilities that include more product terms but could occur from selection rule.

8. Simplify into Sum-of-product form the following Boolean functions F together with the don't-care conditions d:

(a) $F(W, X, Y, Z) = \sum m(0, 1, 3, 5, 7), d(W, X, Y, Z) = \sum m(2, 4, 6)$

Answer:

WX \ YZ	00	01	11	10
00	1	1	1	X
01	X	1	1	X
11	0	0	0	0
10	0	0	0	0

$F = \bar{W}$

(b) $F(A, B, C, D) = \sum m(2, 6, 7, 11, 14, 15), d(A, B, C, D) = \sum m(0, 8, 12, 13)$

Answer:

AB \ CD	00	01	11	10
00	X	0	0	1
01	0	0	1	1
11	X	X	1	1
10	X	0	1	0

$F = BC + \bar{A}C\bar{D} + ACD$

(c) $F(A, B, C, D) = \sum m(2, 7, 9, 10, 15), d(A, B, C, D) = \sum m(3, 6, 14)$

Answer:

AB \ CD	00	01	11	10
00	0	0	X	1
01	0	0	1	X
11	0	0	1	X
10	0	1	0	1

$F = C\bar{D} + BC + A\bar{B}\bar{C}D$